THE UK UNIVERSITY INTEGRATION BEE 2024/2025

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ROUND TWO SOLUTIONS

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Group Round Solutions

1. $\displaystyle\int_0^1 \frac{1}{1+(x+x^2+x^3+x^4+x^5+\dots)^{2024}}\,\mathrm{d}x$

Answer(s):

$\frac{1}{2}$.

Solution. Apply the geometric series. We get

$$\begin{aligned} I &:= \int_0^1 \frac{1}{1+(x+x^2+x^3+x^4+x^5+\ldots)^{2024}}\,\mathrm{d}x \\ &= \int_0^1 \frac{1}{1+\frac{x^{2024}}{(1-x)^{2024}}}\,\mathrm{d}x \\ &= \int_0^1 \frac{(1-x)^{2024}}{(1-x)^{2024}+x^{2024}}\,\mathrm{d}x. \end{aligned}$$

Using King's rule, we have

$$I = \int_0^1 \frac{x^{2024}}{x^{2024}+(1-x)^{2024}}\,\mathrm{d}x,$$

thus $2I = 1$ and $I = \dfrac{1}{2}$. □

2. $\displaystyle\int \frac{\ln(e^x-1)-x}{e^x-1}\,\mathrm{d}x$

Answer(s):

$\dfrac{(\ln(1-e^{-x}))^2}{2}+C,$

$\dfrac{(x-\ln(e^x-1))^2}{2}+C.$

Solution. Interpret x as $\ln(e^x)$, and we have

$$\int \frac{\ln(e^x-1)-x}{e^x-1}\,\mathrm{d}x = \int \frac{e^{-x}\ln(1-e^{-x})}{1-e^{-x}}\,\mathrm{d}x.$$

With $u = 1-e^{-x}$, this becomes just

$$\begin{aligned}\int \frac{\ln(u)}{u}\,\mathrm{d}u &= \frac{(\ln(u))^2}{2}+C\\ &= \frac{(\ln(1-e^{-x}))^2}{2}+C\\ &= \frac{(x-\ln(e^x-1))^2}{2}+C.\end{aligned}$$

This last form can be obtained quickly if one recognizes that $u = \ln(e^x-1)-x$ works too. □

3. $\displaystyle\int_{-\infty}^{\infty} \frac{\cos(x) + x\sin(x)}{x^2 + \cos^2(x)}\,\mathrm{d}x$

Answer(s):

π.

Solution. Dividing both numerator and denominator by $\cos^2(x)$ gives

$$\int_{-\infty}^{\infty} \frac{\cos(x) + x\sin(x)}{x^2 + \cos^2(x)}\,\mathrm{d}x = \int_{-\infty}^{\infty} \frac{\sec(x) + x\sec(x)\tan(x)}{x^2\sec^2(x) + 1}\,\mathrm{d}x.$$

Let $u = x^2\sec^2(x) + 1$. Observe that

$$\begin{aligned}\frac{\mathrm{d}u}{\mathrm{d}x} &= 2x^2\sec^2(x)\tan(x) + 2x\sec^2(x)\\ &= 2x\sec(x)(\sec(x) + x\sec(x)\tan(x))\\ &= \begin{cases} \dfrac{2}{\sqrt{u-1}}(\sec(x) + x\sec(x)\tan(x)), & x \geq 0;\\ -\dfrac{2}{\sqrt{u-1}}(\sec(x) + x\sec(x)\tan(x)), & x \leq 0, \end{cases}\end{aligned}$$

so this substitution works but we need to be careful! The integral becomes

$$\int_1^{\infty} \frac{1}{u\sqrt{u-1}}\,\mathrm{d}u = 2\int_0^{\infty} \frac{1}{1+y^2}\,\mathrm{d}u = \pi.$$

□

4. $\displaystyle\int_0^{\frac{\pi}{2}} \frac{\arctan(\sin(x))}{\sin(x)}\,\mathrm{d}x$

Answer(s):

$\dfrac{\pi}{2}\ln(1+\sqrt{2})$.

Solution. Let

$$I(a) := \int_0^{\frac{\pi}{2}} \frac{\arctan(a\sin(x))}{\sin(x)}\,\mathrm{d}x.$$

Then we have

$$\begin{aligned} I'(a) &= \int_0^{\frac{\pi}{2}} \frac{1}{1+(a\sin(x))^2}\,\mathrm{d}x \\ &= \int_0^{\frac{\pi}{2}} \frac{1}{1+a^2\sin^2(x)}\,\mathrm{d}x \\ &= \int_0^{\frac{\pi}{2}} \frac{\csc^2(x)}{\csc^2(x)+a^2}\,\mathrm{d}x \\ &= \int_0^{\frac{\pi}{2}} \frac{\csc^2(x)}{\cot^2(x)+a^2+1}\,\mathrm{d}x. \end{aligned}$$

Put $u = \cot(x)$. Then we get

$$\begin{aligned} -\int_\infty^0 \frac{1}{u^2+a^2+1}\,\mathrm{d}u = \int_0^\infty \frac{1}{u^2+a^2+1}\,\mathrm{d}u &= \left[\frac{1}{\sqrt{a^2+1}}\tan^{-1}\left(\frac{x}{\sqrt{a^2+1}}\right)\right]_0^\infty \\ &= \frac{\pi}{2\sqrt{a^2+1}}. \end{aligned}$$

It follows

$$I(a) = \int \frac{\pi}{2\sqrt{a^2+1}}\,\mathrm{d}a = \frac{\pi}{2}\sinh^{-1}(a) + C$$

for some constant C. Noting that $I(0) = 0$, it follows $C = 0$, and thus we have

$$\begin{aligned} I(1) &= \frac{\pi}{2}\sinh^{-1}(1) \\ &= \frac{\pi}{2}\ln(1+\sqrt{2}). \end{aligned}$$

□

5. $\displaystyle\int_0^1 \ln\left(\sqrt{1+\sqrt{x}}+\sqrt{1-\sqrt{x}}\right)\,\mathrm{d}x$

Answer(s):

$\dfrac{\ln(2)}{2}+\dfrac{1}{4}$.

Solution. We use and abuse the rules of logarithms to introduce a square in the argument and get

$$\begin{aligned}\int_0^1 \ln\left(\sqrt{1+\sqrt{x}}+\sqrt{1-\sqrt{x}}\right)\,\mathrm{d}x &= \frac{1}{2}\int_0^1 \ln\left(2+2\sqrt{1-x}\right)\,\mathrm{d}x \\ &= \frac{\ln(2)}{2}+\frac{1}{2}\int_0^1 \ln\left(1+\sqrt{1-x}\right)\,\mathrm{d}x.\end{aligned}$$

This last integral is evaluated using $u=\sqrt{1-x}$, and get

$$\begin{aligned}\int_0^1 \ln\left(1+\sqrt{1-x}\right)\,\mathrm{d}x &= 2\int_0^1 u\ln(1+u)\,\mathrm{d}u \\ &= 2\int_0^1 u\ln(1+u)\,\mathrm{d}u \\ &= \frac{1}{2}.\end{aligned}$$

Hence the original integral evaluates to $\dfrac{\ln(2)}{2}+\dfrac{1}{4}$. □

6. $\displaystyle\int_0^e f(e^x)\,\mathrm{d}x$,

where $f(x)$ is the unique value $c \geq 1$ such that $c^c = x$.

Answer(s):

$$\frac{3e^2}{4} - \frac{1}{4}.$$

Solution. Let $x = y\ln(y)$. Then we have

$$\begin{aligned}\int_0^e f(e^x)\,\mathrm{d}x &= \int_1^e f(y^y)(\ln(y)+1)\,\mathrm{d}y \\ &= \int_1^e y\ln(y) + y\,\mathrm{d}y \\ &= \left[\frac{y^2\ln(y)}{2} + \frac{y^2}{4}\right]_1^e \\ &= \frac{3e^2}{4} - \frac{1}{4}.\end{aligned}$$

□

7. $\displaystyle\int_0^\infty \frac{\sin(x)\sin(\sin(x))}{x^2} e^{\cos(x)}\,\mathrm{d}x$

Answer(s):

$\dfrac{\pi(e-1)}{2}$.

Solution. One recognises to complexify the integral, and then the solution practically writes itself:

$$\begin{aligned}
&\int_0^\infty \frac{\sin(x)\sin(\sin(x))}{x^2} e^{\cos(x)}\,\mathrm{d}x \\
&= \operatorname{Im}\left(\int_0^\infty \frac{\sin(x)}{x^2} e^{e^{ix}}\,\mathrm{d}x\right) \\
&= \sum_{n=0}^\infty \frac{1}{n!}\left(\int_0^\infty \frac{\sin(x)\operatorname{Im}(e^{inx})}{x^2}\,\mathrm{d}x\right) \\
&= \sum_{n=1}^\infty \frac{1}{n!}\left(\int_0^\infty \frac{\sin(x)\sin(nx)}{x^2}\,\mathrm{d}x\right) \\
&= \sum_{n=1}^\infty \frac{1}{n!}\left(\int_0^\infty \frac{\cos((n-1)x)-\cos((n+1)x)}{2x^2}\,\mathrm{d}x\right) \\
&= \sum_{n=1}^\infty \frac{1}{n!}\left(\int_0^\infty \int_{n-1}^{n+1} \frac{\sin(yx)}{2x}\,\mathrm{d}y\,\mathrm{d}x\right) \\
&= \sum_{n=1}^\infty \frac{1}{n!}\left(\int_{n-1}^{n+1} \int_0^\infty \frac{\sin(yx)}{2x}\,\mathrm{d}x\,\mathrm{d}y\right) \\
&= \sum_{n=1}^\infty \frac{1}{n!}\left(\int_{n-1}^{n+1} \frac{\pi}{4}\,\mathrm{d}y\right) \\
&= \frac{\pi(e-1)}{2}.
\end{aligned}$$

□

8. $\displaystyle\int_0^1 \operatorname*{argmax}_{y\in\mathbb{R}} \left(x^{\sqrt{1-x^4}y^2+\sqrt{1+x^2}y-\sqrt{1-x^2}y-\sqrt{1-x^8}}\right)\,\mathrm{d}x,$

where $\operatorname*{argmax}_{y\in A}(f(y))$ is the value of $y \in A$ that maximises f

Answer(s):

$\dfrac{\ln(1+\sqrt{2})}{2}-\dfrac{\pi}{4}$,
$\dfrac{\sinh^{-1}(1)}{2}-\dfrac{\pi}{4}$.

Solution. Observe that for a fixed $x \in (0,1)$, the function $x^{\sqrt{1-x^4}y^2+\sqrt{1+x^2}y-\sqrt{1-x^2}y-\sqrt{1-x^8}}$ is maximised when the power is minimal. Hence we have

$$\begin{aligned}&\int_0^1 \operatorname*{argmax}_{y\in\mathbb{R}} \left(x^{\sqrt{1-x^4}y^2+\sqrt{1+x^2}y-\sqrt{1-x^2}y-\sqrt{1-x^8}}\right)\,\mathrm{d}x\\ &\quad=\int_0^1 \operatorname*{argmin}_{y\in\mathbb{R}} \left(\sqrt{1-x^4}y^2+\sqrt{1+x^2}y-\sqrt{1-x^2}y-\sqrt{1-x^8}\right)\mathrm{d}x.\end{aligned}$$

The function to determine a minimum for is (convex) quadratic, hence the minimum occurs at

$$y=\frac{\sqrt{1-x^2}-\sqrt{1+x^2}}{2\sqrt{1-x^4}}=\frac{1}{2\sqrt{1+x^2}}-\frac{1}{2\sqrt{1-x^2}}.$$

Thus the integral is

$$\begin{aligned}\frac{1}{2}\int_0^1 \frac{1}{\sqrt{1+x^2}}-\frac{1}{\sqrt{1-x^2}}\,\mathrm{d}x&=\frac{1}{2}[\sinh^{-1}(x)-\sin^{-1}(x)]_0^1\\ &=\frac{\sinh^{-1}(1)}{2}-\frac{\pi}{4}\\ &=\frac{\ln(1+\sqrt{2})}{2}-\frac{\pi}{4}.\end{aligned}$$

□

9. $\displaystyle\int_0^\infty \frac{\sin(x)}{e^x-1}\,\mathrm{d}x$

Answer(s):

$\dfrac{\pi}{2}\coth(\pi)-\dfrac{1}{2}$.

Solution. We multiply both numerator and denominator by e^{-x} to appeal to geometric series expansion:

$$\int_0^\infty \frac{\sin(x)}{e^x-1}\,\mathrm{d}x=\int_0^\infty \frac{e^{-x}\sin(x)}{1-e^{-x}}\,\mathrm{d}x=\sum_{n=1}^\infty\int_0^\infty \sin(x)e^{-nx}\,\mathrm{d}x.$$

By double integration by parts or by appealing to Laplace transform, we have

$$\int_0^\infty \sin(x)e^{-nx}\,\mathrm{d}x=\frac{1}{n^2+1}.$$

Thus the required value, by appealing to a well-known result of the residue theorem, is

$$\sum_{n=1}^\infty \frac{1}{n^2+1}=\frac{\pi}{2}\coth(\pi)-\frac{1}{2}.$$

Since the competition doesn't require any complex analysis (though knowledge of complex numbers is required), there are methods to evaluate this sum without the residue theorem. The method we use is via the digamma function, $\psi(z)=\dfrac{\Gamma'(z)}{\Gamma(z)}=\dfrac{\mathrm{d}}{\mathrm{d}z}(\ln\Gamma(z))$. The Weierstrass factorisation of the Gamma function states

$$\Gamma(z)=\frac{1}{ze^{\gamma z}}\prod_{n=1}^\infty\left(1+\frac{z}{n}\right)^{-1}e^{\frac{z}{n}}$$

Logarithmically differentiating this, we get an expression for the digamma function

$$\psi(z)=-\frac{1}{z}-\gamma+\sum_{n=1}^\infty\left(\frac{1}{n}-\frac{1}{z+n}\right) \tag{1}$$

Now we use the Gamma reflection formula

$$\Gamma(z)\Gamma(1-z)=\frac{\pi}{\sin(\pi z)}$$

For the digamma function we get a reflection formula upon logarithmically differentiating:

$$\psi(1-z)-\psi(z)=\pi\cot(\pi z)$$

Then we get

$$\begin{aligned}\psi(1-z)-\psi(z)&=\frac{1}{z}+\sum_{k=1}^\infty\frac{1}{k+z}-\frac{1}{k-z}\\&=\frac{1}{z}+\sum_{k=1}^\infty\frac{2z}{z^2-k^2}\\&=\pi\cot(\pi z)\end{aligned}$$

which yields

$$\sum_{k=-\infty}^\infty\frac{1}{z^2-k^2}=\frac{\pi\cot(\pi z)}{z}$$

upon extending the series from $-\infty$ to ∞. Putting in $z=i$ gives the series we needed to evaluate the integral. □

10. $\displaystyle\int_0^1 \frac{1}{1+x}\ln\left(\frac{1+3x}{2-x}\right)\,\mathrm{d}x$

Answer(s):

$\frac{1}{2}(\ln(2))^2$.

Solution. We use the identity

$$\int_a^b (f(g(x)) + f(g^{-1}(x)))f'(x)\,\mathrm{d}x = f(b)^2 - f(a)^2,$$

where f is continuously differentiable and g is an increasing differentiable bijection on $[a,b]$.

Let $f(x) = \ln(1+x)$ and $g(x) = \dfrac{2x}{1+x}$. Clearly g is an increasing differentiable bijection on $[0,1]$, and it can be verified

$$\begin{aligned} f(g(x)) + f(g^{-1}(x)) &= \ln\left(1+\frac{2x}{1+x}\right) + \ln\left(1+\frac{x}{2-x}\right) \\ &= \ln\left(\frac{1+3x}{1+x}\cdot\frac{2}{2-x}\right) \\ &= \ln\left(\frac{1+3x}{2-x}\right) + \ln\left(\frac{2}{1+x}\right). \end{aligned}$$

Now clearly we have

$$\begin{aligned} \int_0^1 \frac{1}{1+x}\ln\left(\frac{2}{1+x}\right)\,\mathrm{d}x &= \int_0^1 \frac{\ln(2)}{1+x} - \frac{\ln(1+x)}{1+x}\,\mathrm{d}x \\ &= \frac{1}{2}(\ln(2))^2. \end{aligned}$$

Therefore,

$$\begin{aligned} \int_0^1 \frac{1}{1+x}\ln\left(\frac{1+3x}{2-x}\right)\,\mathrm{d}x &= (\ln(2))^2 - (\ln(1))^2 - \frac{1}{2}(\ln(2))^2 \\ &= \frac{1}{2}(\ln(2))^2. \end{aligned}$$

Note: In general, one can prove that

$$\int_0^1 \frac{1}{a+x}\ln\left(\frac{a^2+(2a+1)x}{a+1-x}\right)\,\mathrm{d}x = \frac{1}{2}((\ln(a+1))^2 - (\ln(a))^2).$$

□

Crossnumber Round Solutions

Phase 1a: Easy beginnings

ACROSS

1. Differentiate to get $f'(x) = 48x - 25$, integrate to get $f(x) = 24x^2 - 25x + C$. Then $C + \int_{-1}^{1} 24t^2 - 25t + C \, \mathrm{d}t = 3C + 16 = 52$, so $f(0) = C = 12$.

   Alternate solution: Integrating both sides of the equation from -1 to 1, we have

   $$\begin{aligned} 3\int_{-1}^{1} f(t) \, \mathrm{d}t &= \int_{-1}^{1} 24x^2 + 52 \, \mathrm{d}x \\ &= 104 + [8x^3]_{-1}^{1} \\ &= 120 \end{aligned}$$

   so the integral is 40. Subtracting it from the original equation we get $f(0) = 12$ as the constant term of the polynomial.

2. Use the fact that the integrand is even to write it as an integral from 0 to ∞ and let $y = x + \sqrt[b]{a}$. Then we get

   $$\begin{aligned} \int_{-\infty}^{\infty} \frac{1}{\left(|x| + \sqrt[b]{a}\right)\left(\ln\left(|x| + \sqrt[b]{a}\right)\right)^2} \, \mathrm{d}x &= 2\int_{\sqrt[b]{a}}^{\infty} \frac{1}{y(\ln(y))^2} \, \mathrm{d}y \\ &= \frac{2b}{\ln(a)} \\ &= \frac{14}{\ln 12}. \end{aligned}$$

   Thus $a = 12, b = 7$, and these are the only coprime solutions less than 20, so $ab = 84$.

DOWN

3. The first equation is

   $$45\int_0^1 f(x) \, \mathrm{d}x + 25\int_1^2 f(x) \, \mathrm{d}x = 2025,$$

   so subtracting the first equation from the second we get

   $$20\int_0^3 f(x) - 45\int_0^1 f(x) \, \mathrm{d}x = 20\int_1^3 f(x) - 25\int_0^1 f(x) \, \mathrm{d}x = 2520 - 2025 = 495.$$

   Alternate cursed solution: The equations are satisfied with the function (there are many like it, but this one is mine)

   $$f(x) = \begin{cases} 100, & x \in (0,1], \\ -99, & x \in (1,2], \\ \dfrac{995}{4}, & x \in (2,3). \end{cases}$$

   Thus we have

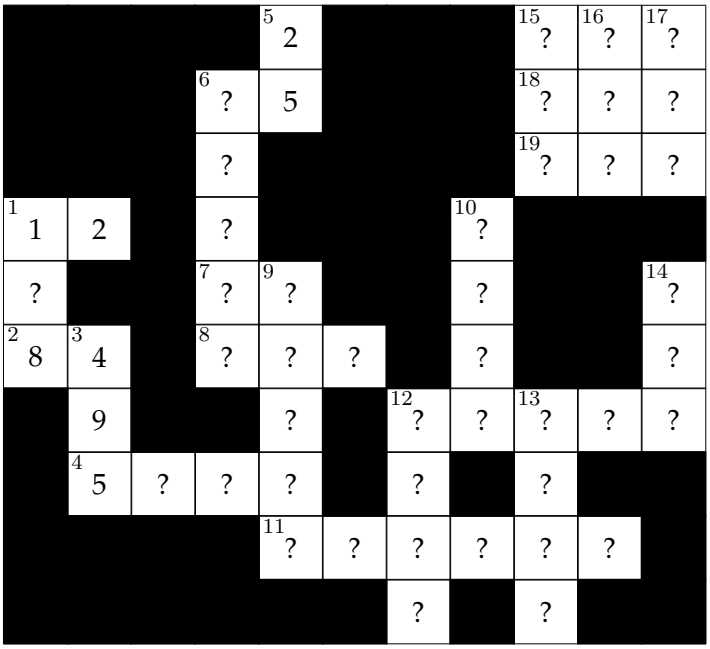
   $$20\int_1^3 f(x) \, \mathrm{d}x - 25\int_0^1 f(x) \, \mathrm{d}x = 20\left(-99 + \frac{995}{4}\right) - 25(100) = 495.$$

5. The integral is

$$\begin{aligned}\int_2^3 \frac{ax+5}{(x-3)^2+1}\,\mathrm{d}x &= \frac{1}{2}\int_2^3 \frac{a(2x-6)+10+6a}{(x-3)^2+1}\,\mathrm{d}x \\ &= -\frac{a\ln(2)}{2} + (5+3a)\frac{\pi}{4}.\end{aligned}$$

Equating gives $a = 25$.

Puzzle state



Phase 1b: Almost independent clues

ACROSS

4. 3 DOWN is known. The only four-digit perfect cube starting with 5 is $18^3 = 5832$.

DOWN

1. Knowing 2 ACROSS, it's easy to see the triple is $(1, 2, 8)$. (This is partially because 8 does not appear as a digit in any cube from 0^3 to 9^3.)

10. 1 ACROSS and 2 ACROSS are known. The integral in consideration is

$$\begin{aligned}\int_0^\pi \left(\sum_{n=1}^{\infty} \frac{\sin(nx)}{n}\right)^2 \mathrm{d}x &= \int_0^\pi \sum_{m=1}^{\infty}\sum_{n=1}^{\infty} \frac{\sin(mx)}{m}\frac{\sin(nx)}{n}\,\mathrm{d}x \\ &= \int_0^\pi \sum_{n=1}^{\infty} \frac{\sin^2(nx)}{n^2}\,\mathrm{d}x \quad \text{(by the orthogonality relations)} \\ &= \sum_{n=1}^{\infty}\frac{1}{n^2}\int_0^\pi \sin^2(nx)\,\mathrm{d}x \\ &= \frac{\pi}{2}\sum_{n=1}^{\infty}\frac{1}{n^2} \\ &= \frac{\pi^3}{12}.\end{aligned}$$

Accordingly, the answer is $12^3 = 1728$.

Alternate solution: The series can be evaluated directly using exponentials. We have

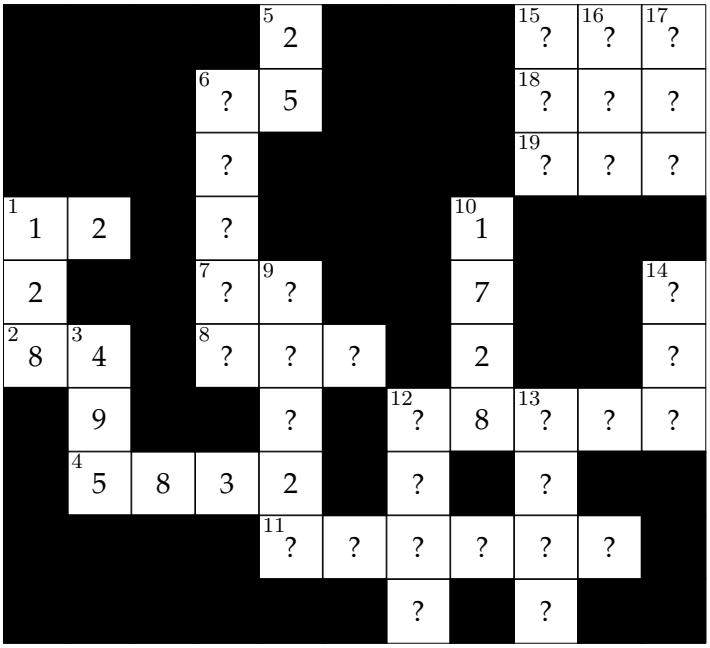
$$\sum_{n=1}^{\infty}\frac{\sin(nx)}{n} = \operatorname{Im}\left(\sum_{n=1}^{\infty}\frac{e^{inx}}{n}\right) = -\operatorname{Im}(\ln(1-e^{ix})) = -\operatorname{Arg}(1-e^{ix}) = \frac{\pi}{2} - \frac{x}{2},$$

where the last equivalence comes from expanding the exponential and using the double angle formula. Therefore the integral is

$$\int_0^\pi \left(\frac{x}{2} - \frac{\pi}{2}\right)^2 \mathrm{d}x = \frac{\pi^3}{12}$$

in the same way we got above.

Puzzle state



At this point, we transition out of the easy phase into the main difficulty of the puzzle. The crux lies in solving 7 ACROSS, 8 ACROSS, 13 DOWN, and to a lesser extent, 11 ACROSS. There is a bit of redundancy and (more or less) two general paths to a solution – this is by design.

Phase 2-1a: Squares, squares everywhere

Phase 2-1 (Phase 2, Route 1) relies on DOWN being able to solve 13 DOWN. In return, ACROSS does not need to apply the concept of simultaneous clues.

ACROSS

7. Knowing the first two digits of 13 DOWN, this is 89 and so 13 DOWN is $89^2 = 7921$.

8. Make the substitution $t = xy$ and $y = y$, then we have

$$\int_0^1 \int_0^1 (xy)^{xy} \, \mathrm{d}x \, \mathrm{d}y = \int_0^1 \int_t^1 t^t \frac{1}{y} \, \mathrm{d}y \, \mathrm{d}t = \int_0^1 t^t(-\ln t) \, \mathrm{d}t.$$

Now $\displaystyle\int_0^1 t^t + t^t \ln t \, \mathrm{d}t = [t^t]_0^1 = 0$, so $\displaystyle\int_0^1 t^t(-\ln t) \, \mathrm{d}t = \int_0^1 t^t \, \mathrm{d}t$. Then $a = b = 1$, so the answer is just $25 + 89 = 114$.

DOWN

13. There's not enough information to do this clue fully, but at least the first two digits can be worked out. There are two ways to perform this integral. The elementary way is to write

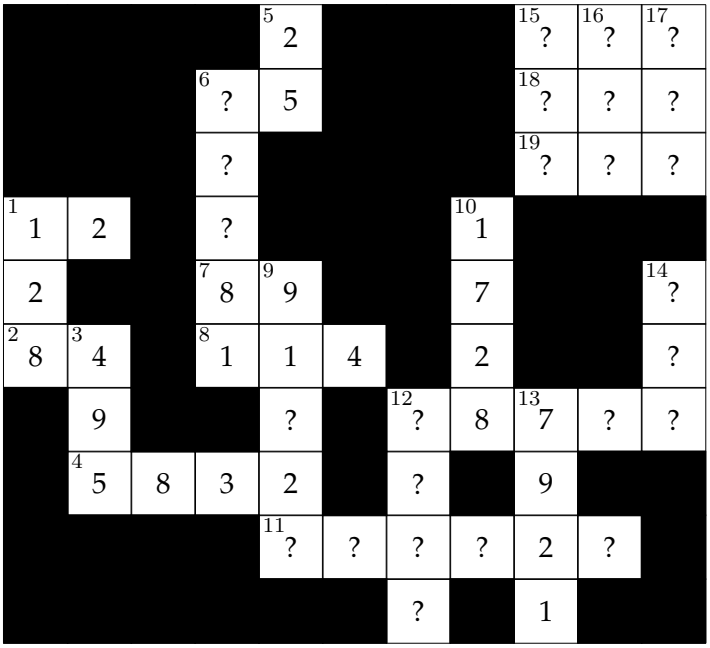
$$\begin{aligned}
\lim_{n\to\infty} \left(\int_{-\infty}^{\infty} \left(e^{-x^2+x}\right)^n \mathrm{d}x \right)^{\frac{1}{n}} &= e^{\frac{1}{4}} \lim_{n\to\infty} \left(\int_{-\infty}^{\infty} \left(e^{-(x-\frac{1}{2})^2}\right)^n \mathrm{d}x \right)^{\frac{1}{n}} \\
&= e^{\frac{1}{4}} \lim_{n\to\infty} \left(\int_{-\infty}^{\infty} \left(e^{-x^2}\right)^n \mathrm{d}x \right)^{\frac{1}{n}} \\
&= e^{\frac{1}{4}} \lim_{n\to\infty} \left(\frac{1}{\sqrt{n}} \int_{-\infty}^{\infty} e^{-x^2} \, \mathrm{d}x \right)^{\frac{1}{n}} \\
&= e^{\frac{1}{4}} \lim_{n\to\infty} \sqrt{\frac{\sqrt[n]{\pi}}{\sqrt[n]{n}}} \\
&= e^{\frac{1}{4}}.
\end{aligned}$$

The big brain way is to use functional analysis and realize the limit is just the supremum norm, i.e.

$$\sup_{x\in\mathbb{R}} |e^{-x^2+x}| = e^{\frac{1}{4}}.$$

Then $b - a = 2$ and $a + b = 16$, and so $a = 7, b = 9$.

Puzzle state



Phase 2-1b: A convex complex

ACROSS

6. It's a Borwein integral with the value $n = 57$, but don't bother solving it because DOWN will finish this clue for you. (Not like you can do it with the information here anyway.)

11. Let $k = \dfrac{a}{b}$. The inequality stated now reads

$$\frac{2}{5}\int_0^1 f(x)\,\mathrm{d}x + \frac{2}{3}\int_0^{\frac{3}{5}} f(x)\,\mathrm{d}x \geq \int_0^k f(x)\,\mathrm{d}x.$$

Guessing a candidate answer is easy. Take $f(x) = x$ and find immediately $k \leq \dfrac{4}{5}$.

Proving it needs more work (though this isn't a proof competition). Consider the Jensen inequality

$$f(x) + f\left(\frac{3}{5}x\right) \geq 2f\left(\frac{4}{5}x\right).$$

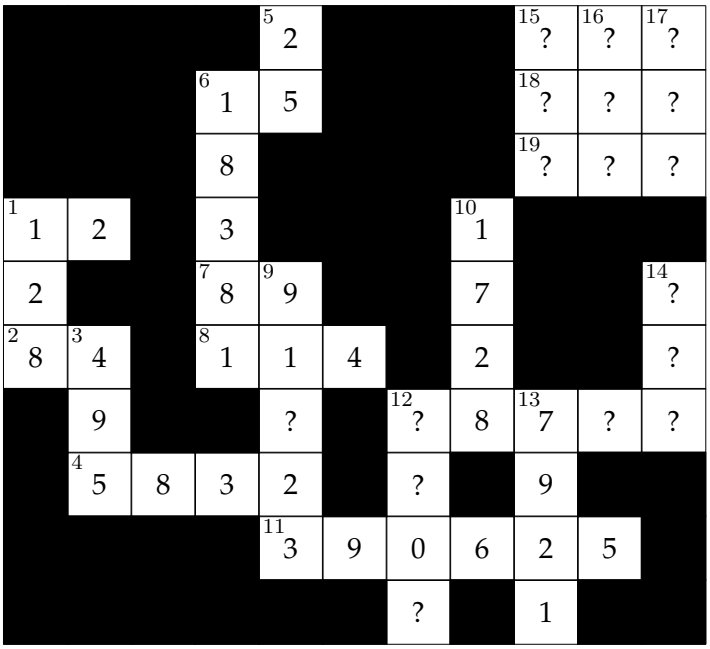
Integrating this (valid because convex functions) and performing linear change in variables we show $k = \dfrac{4}{5}$ works for every continuous convex function, so $k \geq \dfrac{4}{5}$.

Thus $a = 4, b = 5$, and $(b^a)^2 = (5^4)^2 = 390625$.

DOWN

6. After 7 ACROSS and 8 ACROSS are known, palindromic means first two digits are 18, and then just do basic divisibility testing to get 18381.

Puzzle state



Phase 2-2a: Simultaneous clues

Phase 2-2 (Phase 2, Route 2) relies on ACROSS being able to apply partial solving and resolve simultaneous clues. In return, DOWN does not need to solve 13 DOWN's integral.

ACROSS

6. It's a Borwein integral with the value $n = 57$, but don't bother solving it because DOWN will finish this clue for you. (Not like you can do it with the information here anyway.)

8. The most difficult part of this path. Participants need to recognize that *the integral can be fully solved now but the clue can't be*. First, let's solve the integral. Make the substitution $t = xy$ and $y = y$, then we have

$$\int_0^1 \int_0^1 (xy)^{xy} \, \mathrm{d}x \, \mathrm{d}y = \int_0^1 \int_t^1 t^t \frac{1}{y} \, \mathrm{d}y \, \mathrm{d}t = \int_0^1 t^t(-\ln t) \, \mathrm{d}t.$$

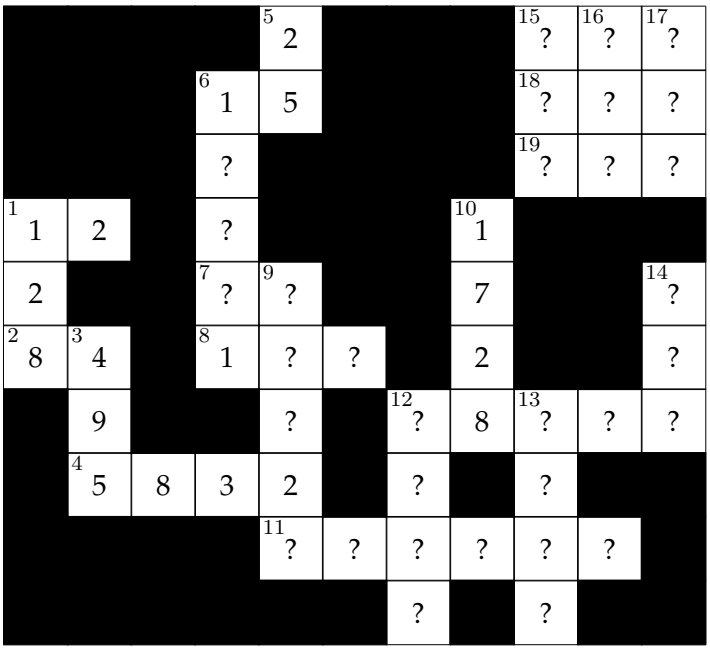
Now $\displaystyle\int_0^1 t^t + t^t \ln t \, \mathrm{d}t = [t^t]_0^1 = 0$, so $\displaystyle\int_0^1 t^t(-\ln t) \, \mathrm{d}t = \int_0^1 t^t \, \mathrm{d}t$. Then $a = b = 1$.

We don't know what (7 ACROSS) is, *but we know it's a two-digit integer*, whereas (7 ACROSS) + (5 DOWN) is a *three-digit integer*. The first digit must therefore be 1.

DOWN

6. After the first digit of 8 ACROSS, the first digit of this must be 1.

Puzzle state



Phase 2-2b: Simultaneous clues, the sequel

ACROSS

11. Let $k = \dfrac{a}{b}$. The inequality stated now reads

$$\frac{2}{5}\int_0^1 f(x)\,\mathrm{d}x + \frac{2}{3}\int_0^{\frac{3}{5}} f(x)\,\mathrm{d}x \geq \int_0^k f(x)\,\mathrm{d}x.$$

Guessing a candidate answer is easy. Take $f(x) = x$ and find immediately $k \leq \dfrac{4}{5}$.

Proving it needs more work (though this isn't a proof competition). Consider the Jensen inequality

$$f(x) + f\left(\frac{3}{5}x\right) \geq 2f\left(\frac{4}{5}x\right).$$

Integrating this (valid because convex functions) and performing linear change in variables we show $k = \dfrac{4}{5}$ works for every continuous convex function, so $k \geq \dfrac{4}{5}$.

Thus $a = 4, b = 5$, and $(b^a)^2 = (5^4)^2 = 390625$.

7. Combine last two digits of 13 DOWN with knowledge that we're looking for a four-digit square. There are three candidates, $39^2 = 1521$, $61^2 = 3721$, and $89^2 = 7921$. Now 8 ACROSS has three digits, so it must be $89^2 = 7921$. This clears 13 DOWN (7921) and 8 ACROSS (114) at the same time too.

DOWN

6. After 7 ACROSS is known, palindromic means second digit is 8, and then just do basic divisibility testing to get 18381.

13. After 11 ACROSS is known, clearly the last digit is 1. There's enough information to solve for this clue fully now.

Solution 1: Your other half does it and you don't need to do anything.

Solution 2: Work out a and b. There are two ways to perform this integral. The elementary way is to write

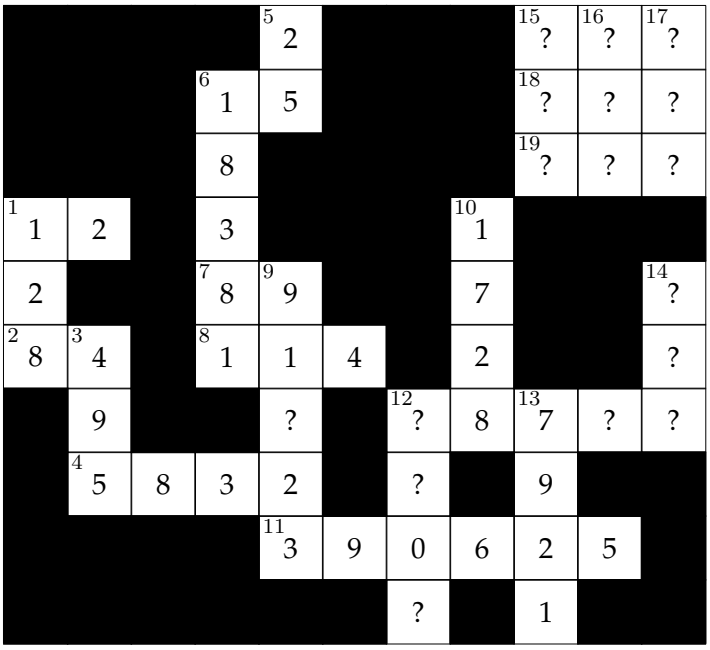
$$\begin{aligned}
\lim_{n\to\infty}\left(\int_{-\infty}^{\infty}\left(e^{-x^2+x}\right)^n \mathrm{d}x\right)^{\frac{1}{n}} &= e^{\frac{1}{4}}\lim_{n\to\infty}\left(\int_{-\infty}^{\infty}\left(e^{-(x-\frac{1}{2})^2}\right)^n \mathrm{d}x\right)^{\frac{1}{n}} \\
&= e^{\frac{1}{4}}\lim_{n\to\infty}\left(\int_{-\infty}^{\infty}\left(e^{-x^2}\right)^n \mathrm{d}x\right)^{\frac{1}{n}} \\
&= e^{\frac{1}{4}}\lim_{n\to\infty}\left(\frac{1}{\sqrt{n}}\int_{-\infty}^{\infty} e^{-x^2}\,\mathrm{d}x\right)^{\frac{1}{n}} \\
&= e^{\frac{1}{4}}\lim_{n\to\infty}\sqrt{\frac{\sqrt[n]{\pi}}{\sqrt[n]{n}}} \\
&= e^{\frac{1}{4}}.
\end{aligned}$$

The big brain way is to use functional analysis and realize the limit is just the supremum norm, i.e.

$$\sup_{x\in\mathbb{R}} |e^{-x^2+x}| = e^{\frac{1}{4}}.$$

Then $b-a=2$ and $a+b=16$, and so $a=7, b=9$.

Puzzle state



Phase 3: The victory lap

ACROSS

12. After 12 DOWN is known, this is clearly 58720.

DOWN

9. After 12 DOWN is known (specifically, the first digit which it shares with 12 ACROSS), 9 DOWN is itself the second-largest. Just estimate. You will get that the middle digit is 4.

12. Now that 2 ACROSS and 7 ACROSS are both known, the integral is

$$\int_0^{173} \lfloor x\lfloor x\rfloor\rfloor - \left\lfloor\sqrt{x\lfloor x\rfloor}\right\rfloor^2 \,\mathrm{d}x.$$

We determine how the bounds split over $[m, m+1)$, $m \in \mathbb{Z}^+$ (for $m = 0$, the integrand is just 0). We have $m \le \sqrt{mx} < \sqrt{m^2+m} < m+1$ so $\left\lfloor\sqrt{mx}\right\rfloor^2 = m$. Then the integrand evaluates as

$$\lfloor mx\rfloor - m^2 = k, \qquad \text{for } m + \frac{k}{m} \le x < m + \frac{k+1}{m}, \quad k = 0, 1, \ldots, m-1,$$

and so

$$\int_m^{m+1} \lfloor x\lfloor x\rfloor\rfloor - \left\lfloor\sqrt{x\lfloor x\rfloor}\right\rfloor^2 \,\mathrm{d}x = \sum_{k=0}^{m-1} \int_{m+\frac{k}{m}}^{m+\frac{k+1}{m}} k \,\mathrm{d}x = \frac{1}{m}\sum_{k=0}^{m-1} k = \frac{m-1}{2}.$$

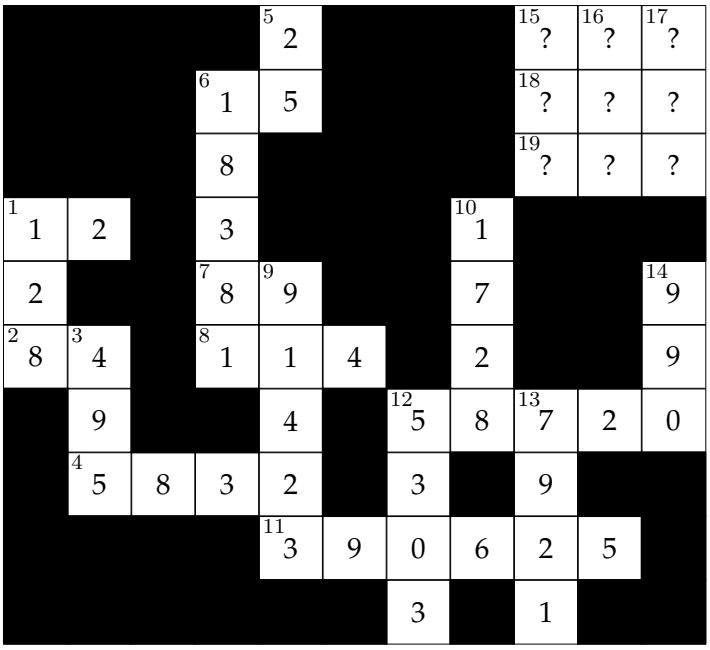
Thus the integral is

$$\sum_{m=0}^{n-1} \int_m^{m+1} \lfloor x\lfloor x\rfloor\rfloor - \left\lfloor\sqrt{x\lfloor x\rfloor}\right\rfloor^2 \,\mathrm{d}x = 0 + \sum_{m=1}^{n-1} \frac{m-1}{2} = \frac{(n-2)(n-1)}{4}.$$

Putting $n = 173$ gives 7353, so the answer is 5303.

14. Once 12 ACROSS is known, the last digit of this answer is 0, so it must be $495 \times 2 = 990$.

Puzzle state



Phase 4: A communications grid

This grid is designed so that only one clue can be handled before a switch to the other half is needed.

DOWN

17. The possible combinations are 389, 479, 569. (578 doesn't apply because it's even.) In any case, the last digit is 9.

15 ?	16 ?	17 ?
18 ?	?	?
19 ?	?	9

ACROSS

19. Only two three-digit squares above 500 end in 9, 529 and 729. Either way, the second-last digit is 2.

15 ?	16 ?	17 ?
18 ?	?	?
19 ?	2	9

DOWN

16. Last digit is 2, so first digit of 15 DOWN is 5.

15 5	16 ?	17 ?
18 ?	?	?
19 ?	2	9

ACROSS

19. 529 is out, so it's 729.

15 5	16 ?	17 ?
18 ?	?	?
19 7	2	9

DOWN

17. Now only 389 works.

15 5	16 ?	17 3
18 ?	?	8
19 7	2	9

ACROSS

15. Only numbers available are 513, 543, 563. Only 563 is prime (and it's sexy with both 557 and 569).

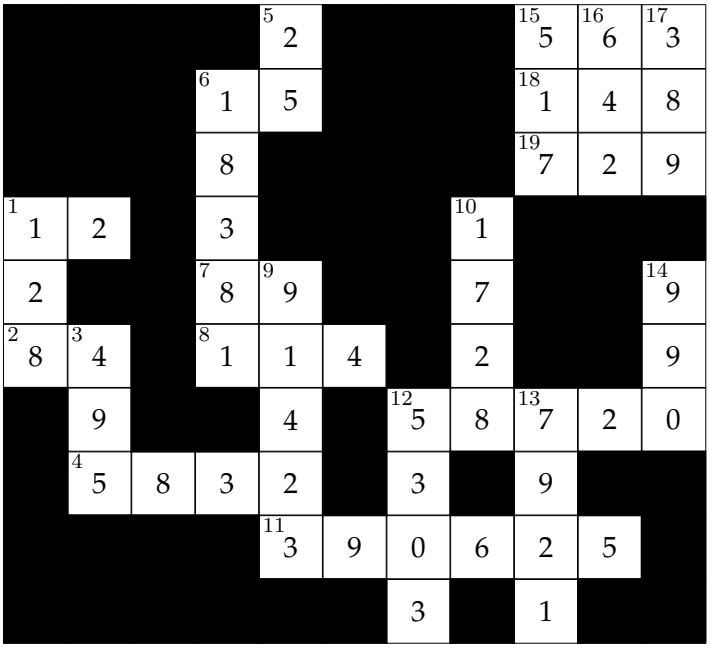
15 5	16 6	17 3
18 ?	?	8
19 7	2	9

DOWN

15. 517 is not prime (divisible by 11) and 547 is. Finish by elimination.

15 5	16 6	17 3
18 1	4	8
19 7	2	9

Puzzle state



Relay Round Solutions

Problems

1. $\displaystyle\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{x^2\cos(x)}{1+e^x}\,\mathrm{d}x$

   Answer(s):

   $\dfrac{\pi^2}{4}-2.$

   Solution. This is an application of the more general result

   $$\int_{-a}^{a} \frac{e(x)}{1+b^{o(x)}}\,\mathrm{d}x = \int_0^a e(x)\,\mathrm{d}x,$$

   where $e(x)$ is an even function and $o(x)$ is an odd function, and $a, b > 0$ are constants. It can be proven by reflection substitution.

   Our integral reduces to a simple integration by parts

   $$\int_0^{\frac{\pi}{2}} x^2\cos(x)\,\mathrm{d}x = \left[(x^2-2)\sin(x)+2x\cos(x)\right]_0^{\frac{\pi}{2}} = \frac{\pi^2}{4}-2.$$

   □

2. $\lim_{n\to\infty} \frac{1}{n} \int_0^1 \left(\frac{1}{n}\right)^x + \left(\frac{2}{n}\right)^x + ... + \left(\frac{n}{n}\right)^x \, \mathrm{d}x$

Answer(s):

$\ln(2)$.

Solution. We recognize a fairly obvious Riemann sum.

$$\begin{aligned}
&\lim_{n\to\infty} \frac{1}{n} \int_0^1 \left(\frac{1}{n}\right)^x + \left(\frac{2}{n}\right)^x + ... + \left(\frac{n}{n}\right)^x \, \mathrm{d}x \\
&= \int_0^1 \lim_{n\to\infty} \frac{1}{n} \sum_{k=1}^{n} \left(\frac{k}{n}\right)^x \, \mathrm{d}x \\
&= \int_0^1 \int_0^1 y^x \, \mathrm{d}y \, \mathrm{d}x \\
&= \int_0^1 \frac{1}{x+1} \, \mathrm{d}x \\
&= \ln(2).
\end{aligned}$$

□

3. $\displaystyle\int_0^1 f(x)\,\mathrm{d}x,$

where σ is the permutation $(3\ 1\ 4)(5\ 9\ 2\ 6)(8\ 7\ 0)$ acting on the digits $0, 1, 2, \ldots, 9$, and $f(x)$ is the function that maps the ith decimal digit x_i of x to $\sigma(x_i)$

Answer(s):

$\dfrac{1}{2}$.

Solution. The intuitive answer is of course $\dfrac{1}{2}$, and it turns out to also be the correct answer. The function is not Riemann integrable, but it is Lebesgue integrable.

Consider the following sequence of functions. For each n, we let

$$f_n(x) = 0.\sigma(x_1)\sigma(x_2)\sigma(x_3)\ldots\sigma(x_n)x_{n+1}x_{n+2}\ldots,$$

where if the decimal representation terminates at the mth digit then we terminate the above accordingly. Observe that since f_n just permutes the graph of $y = x$, we have

$$\int_0^1 f_n(x)\,\mathrm{d}x = \frac{1}{2}.$$

Furthermore, $\lim\limits_{n\to\infty} f_n(x) = f(x)$. By dominated convergence theorem,

$$\int_0^1 f(x)\,\mathrm{d}x = \lim_{n\to\infty}\int_0^1 f_n(x)\,\mathrm{d}x = \frac{1}{2}.$$

□

4. $\displaystyle\int_0^{\pi/2} \cfrac{x^2}{2\cdot 3 - x^2 + \cfrac{2\cdot 3x^2}{4\cdot 5 - x^2 + \cfrac{4\cdot 5x^2}{6\cdot 7 - x^2 + \ddots}}}\,\mathrm{d}x$

Answer(s):

$2G - \dfrac{\pi}{2}$.

Solution. By Taylor expansion of $\sin(x)$ and the very magical Euler's continued fraction formula,

$$\sin(x) = \cfrac{x}{1 - \cfrac{\frac{-x^2}{2\cdot 3}}{1 + \frac{-x^2}{2\cdot 3} - \cfrac{\frac{-x^2}{4\cdot 5}}{1 + \frac{-x^2}{4\cdot 5} - \cfrac{\frac{-x^2}{6\cdot 7}}{1 + \frac{-x^2}{6\cdot 7} - \ddots}}}} = \cfrac{x}{1 + \cfrac{x^2}{2\cdot 3 - x^2 + \cfrac{2\cdot 3x^2}{4\cdot 5 - x^2 + \cfrac{4\cdot 5x^2}{6\cdot 7 - x^2 + \ddots}}}}.$$

(One can do this in reverse, by first reversing the denominator clearing in the integrand.)

Hence the integral is $\displaystyle\int_0^{\pi/2} x\csc(x) - 1\,\mathrm{d}x$. Then we have

$$\begin{aligned}
\int_0^{\pi/2} x\csc(x) - 1\,\mathrm{d}x &= \left[x\ln\left(\tan\left(\frac{x}{2}\right)\right)\right]_0^{\pi/2} - \int_0^{\frac{\pi}{2}} \ln\left(\tan\left(\frac{x}{2}\right)\right)\,\mathrm{d}x - \frac{\pi}{2} \\
&= -2\int_0^1 \frac{\ln(u)}{1+u^2}\,\mathrm{d}u - \frac{\pi}{2} \\
&= -2\sum_{n=0}^{\infty}(-1)^n \int_0^1 u^{2n}\ln(u)\,\mathrm{d}u - \frac{\pi}{2} \\
&= 2\sum_{n=0}^{\infty}(-1)^n \int_0^{\infty} e^{-(2n+1)t} t\,\mathrm{d}t - \frac{\pi}{2} \\
&= 2\sum_{n=0}^{\infty}\frac{(-1)^n}{(2n+1)^2} - \frac{\pi}{2} \\
&= 2G - \frac{\pi}{2}.
\end{aligned}$$

□

5. $\displaystyle\int_{-\infty}^{\infty} \frac{1}{1+(x+\frac{1}{x})^2}\,\mathrm{d}x$

Answer(s):

$\dfrac{\pi}{\sqrt{5}}$.

Solution. We apply Glassers' Master Theorem (methods such as factorization of the denominator also work, we leave this to the reader). We have

$$\begin{aligned}\int_{-\infty}^{\infty} \frac{1}{1+(x+\frac{1}{x})^2}\,\mathrm{d}x &= \int_{-\infty}^{\infty} \frac{1}{5+(x-\frac{1}{x})^2}\,\mathrm{d}x \\ &= \int_{-\infty}^{\infty} \frac{1}{5+x^2}\,\mathrm{d}x \\ &= \frac{\pi}{\sqrt{5}}.\end{aligned}$$

□

6. $\lim_{n\to\infty} n \int_{-\frac{1}{20}}^{\frac{1}{24}} \left(1 - |x| + |x|^3\right)^n \, \mathrm{d}x$

Answer(s):

2.

Solution. Let $y = nx$. Then the integral is

$$\lim_{n\to\infty} n \int_{-\frac{1}{20}}^{\frac{1}{24}} \left(1 - |x| + |x|^3\right)^n \, \mathrm{d}x = \lim_{n\to\infty} \int_{-\frac{n}{20}}^{\frac{n}{24}} \left(1 - \left|\frac{y}{n}\right| + \left|\frac{y}{n}\right|^3\right)^n \, \mathrm{d}y.$$

The physicist would say to ignore the last term and take the integral over $\mathbb{R}$ to pass the limit in. The goated mathematician would mald at the audacity. So we consider the expansion

$$\begin{aligned}
&\lim_{n\to\infty} \int_{-\frac{n}{20}}^{\frac{n}{24}} \left(1 - \left|\frac{y}{n}\right| + \left|\frac{y}{n}\right|^3\right)^n \, \mathrm{d}y \\
&\quad = \lim_{n\to\infty} \int_{-\frac{n}{20}}^{\frac{n}{24}} \left(1 - \left|\frac{y}{n}\right|\right)^n \, \mathrm{d}y + \lim_{n\to\infty} \sum_{k=1}^{n} \binom{n}{k} \int_{-\frac{n}{20}}^{\frac{n}{24}} \left(1 - \left|\frac{y}{n}\right|\right)^{n-k} \left|\frac{y}{n}\right|^{3k} \, \mathrm{d}y.
\end{aligned}$$

The first limit can be dealt with dominated convergence as

$$\begin{aligned}
&\lim_{n\to\infty} \int_{-\frac{n}{20}}^{\frac{n}{24}} \left(1 - \left|\frac{y}{n}\right|\right)^n \, \mathrm{d}y \\
&\quad = \lim_{n\to\infty} \int_{-\infty}^{\infty} \chi_{\left(-\frac{n}{20}, \frac{n}{24}\right)} \left(1 - \left|\frac{y}{n}\right|\right)^n \, \mathrm{d}y \\
&\quad = \int_{-\infty}^{\infty} \lim_{n\to\infty} \chi_{\left(-\frac{n}{20}, \frac{n}{24}\right)} \left(1 - \left|\frac{y}{n}\right|\right)^n \, \mathrm{d}y \\
&\quad = \int_{-\infty}^{\infty} e^{-|y|} \, \mathrm{d}y \\
&\quad = 2,
\end{aligned}$$

where χ represents the indicator function. A similar trick applies in the second limit, but with the crucial observation that in the integration limits, $0 \leq \left(1 - \left|\frac{y}{n}\right|\right)^{n-k} \leq 1$:

$$\begin{aligned}
&\lim_{n\to\infty} \sum_{k=1}^{n} \binom{n}{k} \int_{-\frac{n}{20}}^{\frac{n}{24}} \left(1 - \left|\frac{y}{n}\right|\right)^{n-k} \left|\frac{y}{n}\right|^{3k} \, \mathrm{d}y \\
&\quad = \lim_{n\to\infty} \int_{-\infty}^{\infty} \chi_{\left(-\frac{n}{20}, \frac{n}{24}\right)} \sum_{k=1}^{n} \binom{n}{k} \left(1 - \left|\frac{y}{n}\right|\right)^{n-k} \left|\frac{y}{n}\right|^{3k} \, \mathrm{d}y \\
&\quad \leq \lim_{n\to\infty} \int_{-\infty}^{\infty} \chi_{\left(-\frac{n}{20}, \frac{n}{24}\right)} \sum_{k=1}^{n} \binom{n}{k} \left|\frac{y}{n}\right|^{3k} \, \mathrm{d}y \\
&\quad = \lim_{n\to\infty} \int_{-\infty}^{\infty} \chi_{\left(-\frac{n}{20}, \frac{n}{24}\right)} \left(\left(1 + \left|\frac{y}{n}\right|^3\right)^n - 1\right) \, \mathrm{d}y.
\end{aligned}$$

The limit

$$\lim_{n\to\infty} \left(1 + \left|\frac{y}{n}\right|^3\right)^n = 1$$

holds, so by dominated convergence theorem we may pass the limit in and get

$$\int_{-\infty}^{\infty} \lim_{n\to\infty} \chi_{\left(-\frac{n}{20},\frac{n}{24}\right)} \left(\left(1+\left|\frac{y}{n}\right|^3\right)^n - 1\right) \, \mathrm{d}y = \int_{-\infty}^{\infty} 0 \, \mathrm{d}y = 0.$$

Squeeze theorem shows the second limit is thus 0, and the answer is 2. □

7. $\int_1^2 (\sqrt[x^2]{x-1})^{\mathrm{d}x}$,

where $\int_a^b f(x)^{\mathrm{d}x}$ is the product integral, defined analogously to the Riemann integral with product in place of sum, i.e. $\lim_{\Delta x_i \to 0} \prod_{i=1}^{n} f(x_i)^{\Delta x_i}$ where $a = x_0 < x_1 < \cdots < x_n = b$ and $\Delta x_i = x_i - x_{i-1}$

Answer(s):

$\frac{1}{2}$.

Solution. We observe that

$$\ln\left(\prod_{i=1}^{n} f(x_i)^{\Delta x_i}\right) = \sum_{i=1}^{n} \Delta x_i \ln f(x_i),$$

and therefore (if the limits exist),

$$\ln\left(\int_a^b f(x)^{dx}\right) = \int_a^b \ln(f(x))\,\mathrm{d}x.$$

Hence we have

$$\begin{aligned}\int_1^2 (\sqrt[x^2]{x-1})^{\mathrm{d}x} &= \exp\left(\int_1^2 \ln(\sqrt[x^2]{x-1})\,\mathrm{d}x\right)\\ &= \exp\left(\int_1^2 \frac{\ln(x-1)}{x^2}\,\mathrm{d}x\right).\end{aligned}$$

This integral can be dealt with by parts, as the indefinite integral is

$$\begin{aligned}\int \frac{\ln(x-1)}{x^2}\,\mathrm{d}x &= -\frac{\ln(x-1)}{x} + \int \frac{1}{x(x-1)}\,\mathrm{d}x\\ &= \ln(x-1) - \frac{\ln(x-1)}{x} - \ln(x) + C.\end{aligned}$$

It follows that

$$\exp\left(\int_1^2 \frac{\ln(x-1)}{x^2}\,\mathrm{d}x\right) = \exp\left(-\ln(2)\right) = \frac{1}{2}.$$

□

8. $\displaystyle\int_0^{\frac{\pi}{4}} \sum_{n=2}^{\infty} \left(\frac{1}{\sqrt{n-\sec^2(x)}} - \frac{1}{\sqrt{n-\tan^2(x)}} \right) \mathrm{d}x$

Answer(s):

$\dfrac{\pi}{2\sqrt{2}}$.

Solution. Observe that $\sec^2(x) = \tan^2(x) + 1$. Hence the sum is

$$\sum_{n=2}^{\infty} \left(\frac{1}{\sqrt{n-1-\tan^2(x)}} - \frac{1}{\sqrt{n-\tan^2(x)}} \right) = \frac{1}{\sqrt{1-\tan^2(x)}}.$$

Using trigonometric substitutions, the integral becomes

$$\begin{aligned}
\int_0^{\frac{\pi}{4}} \frac{1}{\sqrt{1-\tan^2(x)}}\,\mathrm{d}x &= \int_0^1 \frac{1}{(1+u^2)\sqrt{1-u^2}}\,\mathrm{d}u \\
&= \int_0^{\frac{\pi}{2}} \frac{1}{1+\sin^2(y)}\,\mathrm{d}y \\
&= \int_0^{\frac{\pi}{2}} \frac{\sec^2(y)}{1+2\tan^2(y)}\,\mathrm{d}y \\
&= \frac{1}{\sqrt{2}} \int_0^{\frac{\pi}{2}} \frac{\sec^2(t)}{1+\tan^2(t)}\,\mathrm{d}t \\
&= \frac{\pi}{2\sqrt{2}}.
\end{aligned}$$

□

9. $\lim_{N\to\infty} \int_0^1 (1-x) \prod_{n=0}^{N} \left(x^{2^n} + \frac{\left\{\sinh^2(x^{2^n})\right\}}{\left\{\cosh^2(x^{2^n})\right\}} \right) \, \mathrm{d}x$

Answer(s):

1.

Solution. From $\cosh^2(x) - \sinh^2(x) = 1$ we get that $\left\{\sinh^2(x^{2^n})\right\} = \left\{\cosh^2(x^{2^n})\right\}$. Then the integral becomes

$$\begin{aligned}
&\lim_{N\to\infty} \int_0^1 (1-x) \prod_{n=0}^{N} \left(x^{2^n} + \frac{\left\{\sinh^2(x^{2^n})\right\}}{\left\{\cosh^2(x^{2^n})\right\}} \right) \, \mathrm{d}x \\
&= \lim_{N\to\infty} \int_0^1 (1-x) \prod_{n=0}^{N} \left(1 + x^{2^n} \right) \, \mathrm{d}x \\
&= \lim_{N\to\infty} \int_0^1 1 - x^{2^{N+1}} \, \mathrm{d}x \\
&= 1.
\end{aligned}$$

□

10. $\displaystyle\int_{-\infty}^{\infty} \frac{e^{i\theta r} - 1 - i\theta \sin(r)}{|r|^{1+\alpha}}$ for $\alpha \in (0,1) \cup (1,2)$ and $\theta \in \mathbb{R}$,

where i is the imaginary unit

Answer(s):

$2\Gamma(-\alpha)\cos\left(\frac{\pi}{2}\alpha\right)|\theta|^{\alpha}$.

Solution. The Lévy-Khintchine representation of the characteristic function of a symmetric α-stable distribution $S_\alpha(c,0,0)$, $\alpha \in (0,2)$, can be written as

$$\exp\left(\int_{-\infty}^{\infty}(e^{i\theta r} - 1 - i\theta\sin(r)) \cdot \frac{c^{\alpha}}{-2\Gamma(-\alpha)\cos\left(\frac{\pi}{2}\alpha\right)|r|^{1+\alpha}}\right).$$

The same characteristic function also has representation

$$\exp\left(-c^{\alpha}|\theta|^{\alpha}\right), \qquad \alpha \in (0,1) \cup (1,2).$$

Equating and rearranging gives

$$\int_{-\infty}^{\infty}(e^{i\theta r} - 1 - i\theta\sin(r)) \cdot \frac{1}{|r|^{1+\alpha}} = 2\Gamma(-\alpha)\cos\left(\frac{\pi}{2}\alpha\right)|\theta|^{\alpha}.$$

□

11. $\displaystyle\int_0^1 \tan^{-1}\left(\frac{\tan^{-1}(\frac{x}{1-x})}{\tan^{-1}(\frac{1-x}{x})}\right)\,\mathrm{d}x$

Answer(s):

$\dfrac{\pi}{4}$.

Solution. We apply King's rule to obtain

$$I := \int_0^1 \tan^{-1}\left(\frac{\tan^{-1}(\frac{x}{1-x})}{\tan^{-1}(\frac{1-x}{x})}\right)\,\mathrm{d}x = \int_0^1 \tan^{-1}\left(\frac{\tan^{-1}(\frac{1-x}{x})}{\tan^{-1}(\frac{x}{1-x})}\right)\,\mathrm{d}x.$$

Applying the identity $\tan^{-1}(x) + \tan^{-1}\left(\dfrac{1}{x}\right) = \dfrac{\pi}{2}$, we get $2I = \dfrac{\pi}{2}$, so $I = \dfrac{\pi}{4}$. □

12. $\displaystyle\int \frac{8+\frac{4+\frac{2+\frac{1}{\sqrt{x}}}{\sqrt{x+\sqrt{x}}}}{\sqrt{x+\sqrt{x+\sqrt{x}}}}}{\sqrt{x+\sqrt{x+\sqrt{x+\sqrt{x}}}}}\,\mathrm{d}x$

Answer(s):

$$16\sqrt{x+\sqrt{x+\sqrt{x+\sqrt{x}}}}+C.$$

Solution. Observe that

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}x}\sqrt{x+\sqrt{x+\sqrt{x+\sqrt{x}}}} &= \frac{1+\frac{\mathrm{d}}{\mathrm{d}x}\sqrt{x+\sqrt{x+\sqrt{x}}}}{2\sqrt{x+\sqrt{x+\sqrt{x+\sqrt{x}}}}} \\
&= \frac{1+\frac{1+\frac{\mathrm{d}}{\mathrm{d}x}\sqrt{x+\sqrt{x}}}{2\sqrt{x+\sqrt{x+\sqrt{x}}}}}{2\sqrt{x+\sqrt{x+\sqrt{x+\sqrt{x}}}}} \\
&= \frac{1+\frac{1+\frac{1+\frac{\mathrm{d}}{\mathrm{d}x}\sqrt{x}}{2\sqrt{x+\sqrt{x}}}}{2\sqrt{x+\sqrt{x+\sqrt{x}}}}}{2\sqrt{x+\sqrt{x+\sqrt{x+\sqrt{x}}}}} \\
&= \frac{1+\frac{1+\frac{1+\frac{1}{2\sqrt{x}}}{2\sqrt{x+\sqrt{x}}}}{2\sqrt{x+\sqrt{x+\sqrt{x}}}}}{2\sqrt{x+\sqrt{x+\sqrt{x+\sqrt{x}}}}} \\
&= \frac{1}{16}\cdot\frac{8+\frac{4+\frac{2+\frac{1}{\sqrt{x}}}{\sqrt{x+\sqrt{x}}}}{\sqrt{x+\sqrt{x+\sqrt{x}}}}}{\sqrt{x+\sqrt{x+\sqrt{x+\sqrt{x}}}}}.
\end{aligned}$$

Thus the answer is $16\sqrt{x+\sqrt{x+\sqrt{x+\sqrt{x}}}}+C$. □

13. $\displaystyle\int_{\frac{1}{4}}^{\frac{1}{2}} (1+\ln(x))\ln(1+x^x)\,\mathrm{d}x$

Answer(s):

0.

Solution. Let $y = x^x$. Then $\dfrac{1}{y}\,\mathrm{d}y = (1+\ln(x))\,\mathrm{d}x$ and

$$\int_{\frac{1}{4}}^{\frac{1}{2}} (1+\ln(x))\ln(1+x^x)\,\mathrm{d}x = \int_{\frac{1}{\sqrt{2}}}^{\frac{1}{\sqrt{2}}} \frac{\ln(1+y)}{y}\,\mathrm{d}y = 0,$$

because the bounds are equal. □

14. $\displaystyle\int_0^{2025} |1-|2-|3-\cdots-|2025-x|\ldots|||\,\mathrm{d}x$

Answer(s):

$$\frac{2025}{2}.$$

Solution. Let $f(x) = |1-|2-|3-\cdots-|2025-x|\ldots|||$. The key to this problem lies in the following three observations:

(a) $f(x)$ takes only values 0 or 1 at integer values $0, 1, 2, \ldots, 2025$.
We consider a general form $|n-1-|n-x||$ and observe how these absolute values evaluate. In the case $x = 0$ this evaluates to 1, and in the case $x = 1$ this evaluates to 0. Iteratively, the result follows for $x = 0, 1$.
Now we consider in general some $k \in [2, 2025]$. It is enough to show that we arrive at $|n-0|$ or $|n-1|$ at some point of the computation. Observe that

$$\begin{aligned} |2025-k| &= 2025-k, \\ |2024-(2025-k)| &= k-1, \\ |2023-(k-1)| &= 2025-(k+1), \\ |2022-(2025-(k+1))| &= k-2, \end{aligned}$$

and so on. It is clear by simple counting that we must eventually reach $|n-0|$ or $|n-1|$.

(b) The values alternate at each successive integer.
Clear from computation argument above (just consider k and $k+1$).

(c) The function is linear between any two successive integers.
Since the constants are all integers, bifurcations can only happen at integer points.

Then the function is a spike function alternating between 0 and 1 at integer points. Accordingly,

$$\int_0^{2025} |1-|2-|3-\cdots-|2025-x|\ldots|||\,\mathrm{d}x = \frac{2025}{2}.$$

□

15. $\displaystyle\int_0^1 \frac{\sin^2(\frac{\pi}{2}x)}{\sqrt{x-x^2}}\,\mathrm{d}x$

Answer(s):

$\dfrac{\pi}{2}$.

Solution. We apply King's rule and get

$$I := \int_0^1 \frac{\sin^2(\frac{\pi}{2}x)}{\sqrt{x-x^2}}\,\mathrm{d}x = \int_0^1 \frac{\sin^2(\frac{\pi}{2}-\frac{\pi}{2}x)}{\sqrt{x-x^2}}\,\mathrm{d}x = \int_0^1 \frac{\cos^2(\frac{\pi}{2}x)}{\sqrt{x-x^2}}\,\mathrm{d}x.$$

Hence,

$$\begin{aligned} 2I &= \int_0^1 \frac{1}{\sqrt{x-x^2}}\,\mathrm{d}x \\ &= \int_0^1 \frac{1}{\sqrt{\frac{1}{4}-\left(x-\frac{1}{2}\right)^2}}\,\mathrm{d}x \\ &= \int_{-\frac{1}{2}}^{\frac{1}{2}} \frac{1}{\sqrt{\frac{1}{4}-y^2}}\,\mathrm{d}y \\ &= [\sin^{-1}(2y)]_{-\frac{1}{2}}^{\frac{1}{2}} \\ &= \pi. \end{aligned}$$

Thus $I = \dfrac{\pi}{2}$. □

16. $\displaystyle\int_{\pi}^{3\pi} \max(\min(x, \sin(x)), \min(x, \cos(x)))\ \mathrm{d}x$

Answer(s):

$2\sqrt{2}$.

Solution. Since the limits of integration are $[\pi, 3\pi]$ the scary minima reduce to

$$\begin{aligned}
&\int_{\pi}^{3\pi} \max(\min(x, \sin(x)), \min(x, \cos(x)))\ \mathrm{d}x \\
&\quad= \int_{\pi}^{3\pi} \max(\sin(x), \cos(x))\ \mathrm{d}x \\
&\quad= \int_{\frac{\pi}{4}}^{\frac{9\pi}{4}} \max(\sin(x), \cos(x))\ \mathrm{d}x \\
&\quad= \int_{\frac{\pi}{4}}^{\frac{5\pi}{4}} \sin(x)\ \mathrm{d}x + \int_{\frac{5\pi}{4}}^{\frac{9\pi}{4}} \cos(x)\ \mathrm{d}x \\
&\quad= [-\cos(x)]_{\frac{\pi}{4}}^{\frac{5\pi}{4}} + [\sin(x)]_{\frac{5\pi}{4}}^{\frac{9\pi}{4}} \\
&\quad= 2\sqrt{2}.
\end{aligned}$$

□

17. $\displaystyle\int_1^\infty \frac{x}{\lfloor x \rfloor^4 + 4}\,\mathrm{d}x$

Answer(s):

$\dfrac{5+\pi\coth(\pi)}{16}$.

Solution. As usual, we split the bounds of this integral:

$$\int_1^\infty \frac{x}{\lfloor x \rfloor^4 + 4}\,\mathrm{d}x = \sum_{n=1}^\infty \int_n^{n+1} \frac{x}{n^4+4}\,\mathrm{d}x = \frac{1}{2}\sum_{n=1}^\infty \frac{2n+1}{n^4+4}.$$

To deal with the sum, we appeal to Sophie Germain's identity, and write

$$\begin{aligned}\sum_{n=1}^\infty \frac{2n+1}{n^4+4} &= \sum_{n=1}^\infty \frac{2n+1}{(n^2+2n+2)(n^2-2n+2)} \\ &= \frac{1}{8}\sum_{n=1}^\infty \left(\frac{n-2}{n^2+2n+2} - \frac{n-6}{n^2-2n+2}\right) \\ &= \frac{1}{8}\sum_{n=1}^\infty \left(\frac{(n+2)-4}{(n+2)^2-2(n+2)+2} - \frac{n-4}{n^2-2n+2}\right) + \frac{1}{4}\sum_{n=1}^\infty \frac{1}{n^2-2n+2}.\end{aligned}$$

The first sum telescopes and evaluates to

$$-\frac{1-4}{1^2-2(1)+2} - \frac{2-4}{2^2-2(2)+2} = 4.$$

We evaluate the second sum by appealing to a well-known result of the residue theorem (see group round solutions for an alternative derivation):

$$\sum_{n=1}^\infty \frac{1}{n^2-2n+2} = \sum_{n=0}^\infty \frac{1}{n^2+1} = \frac{\pi}{2}\coth(\pi) + \frac{1}{2}.$$

Hence the required answer is

$$\frac{1}{2}\left(\frac{1}{8}\cdot 4 + \frac{1}{4}\cdot\left(\frac{\pi}{2}\coth(\pi) + \frac{1}{2}\right)\right) = \frac{5+\pi\coth(\pi)}{16}.$$

□

18. $\displaystyle\int_{-1}^{1} \ln\left(\frac{1+x}{1-x}\right) \ln(1+e^x)\,\mathrm{d}x$

Answer(s):

1.

Solution. Perform a reflection substitution. We have

$$\begin{aligned} I &:= \int_{-1}^{1} \ln\left(\frac{1+x}{1-x}\right) \ln(1+e^x)\,\mathrm{d}x \\ &= \int_{-1}^{1} \ln\left(\frac{1-x}{1+x}\right) \ln(1+e^{-x})\,\mathrm{d}x \\ &= -\int_{-1}^{1} \ln\left(\frac{1+x}{1-x}\right) (\ln(1+e^x)-x)\,\mathrm{d}x. \end{aligned}$$

Thus we get

$$\begin{aligned} 2I &= \int_{-1}^{1} x \ln\left(\frac{1+x}{1-x}\right)\,\mathrm{d}x \\ &= \int_{-1}^{1} x\ln(1+x) - x\ln(1-x)\,\mathrm{d}x \\ &= \frac{1}{2}\left[(x^2-1)\ln\left(\frac{1+x}{1-x}\right) + 2x\right]_{-1}^{1} \\ &= 2. \end{aligned}$$

So $I = 1$. □

19. $\displaystyle\int_{\frac{1}{e}}^{\infty} \frac{\sqrt{\ln(x)+1}}{x^2}\,\mathrm{d}x$

Answer(s):

$e\Gamma\left(\frac{3}{2}\right)$,

$\frac{e\sqrt{\pi}}{2}$.

Solution. Let $y = \ln(x)+1$. Then we have

$$\begin{aligned}\int_{\frac{1}{e}}^{\infty} \frac{\sqrt{\ln(x)+1}}{x^2}\,\mathrm{d}x &= \int_0^{\infty} e^{1-y}\sqrt{y}\,\mathrm{d}y \\ &= e\Gamma\left(\frac{3}{2}\right) = \frac{e\sqrt{\pi}}{2}.\end{aligned}$$

An alternative solution is the following (but works out essentially the same). Inspired by the lower bound $\frac{1}{e}$. Notice that $1 = \ln(e)$ so

$$\int_{\frac{1}{e}}^{\infty} \frac{\sqrt{\ln(x)+1}}{x^2}\,\mathrm{d}x = \int_{\frac{1}{e}}^{\infty} \frac{\sqrt{\ln(xe)}}{x^2}\,\mathrm{d}x = e\int_1^{\infty} \frac{\sqrt{\ln(u)}}{u^2}\,\mathrm{d}u$$

where in the last line we let $u = xe$. Now we let $y = \ln(u)$ which gives us

$$\begin{aligned}e\int_1^{\infty} \frac{\sqrt{\ln(u)}}{u^2}\,\mathrm{d}u &= \int_0^{\infty} e^{1-y}\sqrt{y}\,\mathrm{d}y \\ &= e\Gamma\left(\frac{3}{2}\right) = \frac{e\sqrt{\pi}}{2}.\end{aligned}$$

as we had before. □

20. $\displaystyle\int \frac{(x^2+x+1)(x^3-x+1)}{x^5+1}\,\mathrm{d}x$

Answer(s):

$x+\dfrac{1}{5}\ln(x^5+1)+C.$

Solution. Expanding the numerator and splitting the fraction gives

$$\begin{aligned}\int \frac{(x^2+x+1)(x^3-x+1)}{x^5+1}\,\mathrm{d}x &= \int \frac{x^5+x^4+1}{x^5+1}\,\mathrm{d}x\\ &= \int 1+\frac{x^4}{x^5+1}\,\mathrm{d}x\\ &= x+\frac{1}{5}\ln(x^5+1)+C.\end{aligned}$$

□